

Osteoporosis: The Book I Wish I'd Had When I Was Diagnosed

Companion materials for the audio edition

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Every table, illustration, figure, glossary term and source from the book, in one printable document, along with the 30-day plan as a checklist.

This document accompanies the audio edition. It is also available at mybone.health/listen, so you can reach it from any device or listening app. Educational information only, not medical advice. Talk with your own physician or healthcare provider about what is right for you.

The exercises, drawn

These are the five strength exercises from Chapter 7, plus the safe lifting position from Chapter 11. Check with your own clinician or a physical therapist before you start, especially if you've had a fracture.

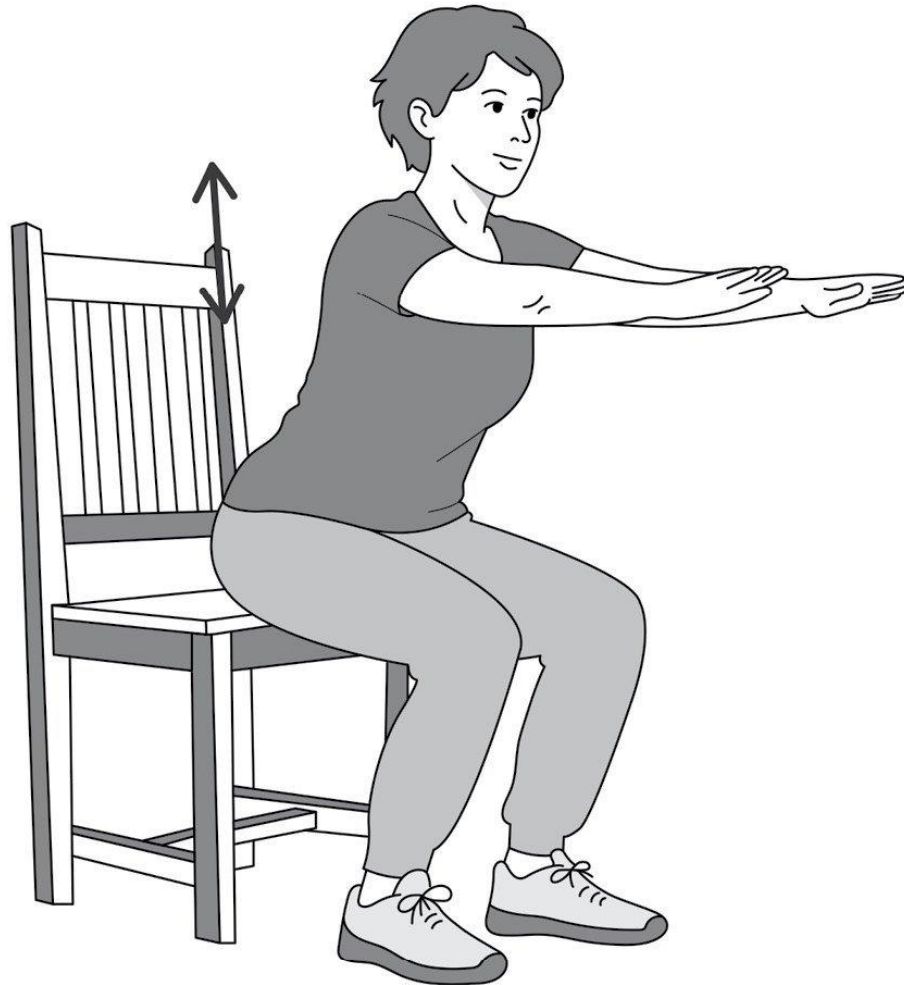


Figure 7.1. Chair squats build leg strength while loading the hip and spine.

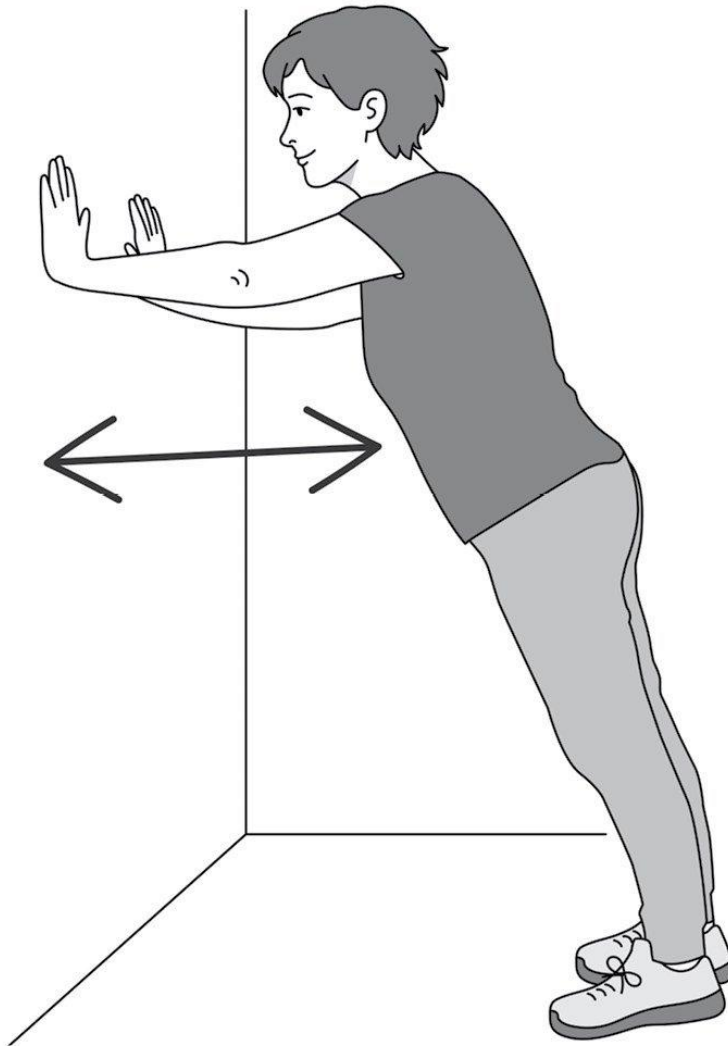


Figure 7.2. Wall push-ups safely strengthen the upper body and load the wrists.

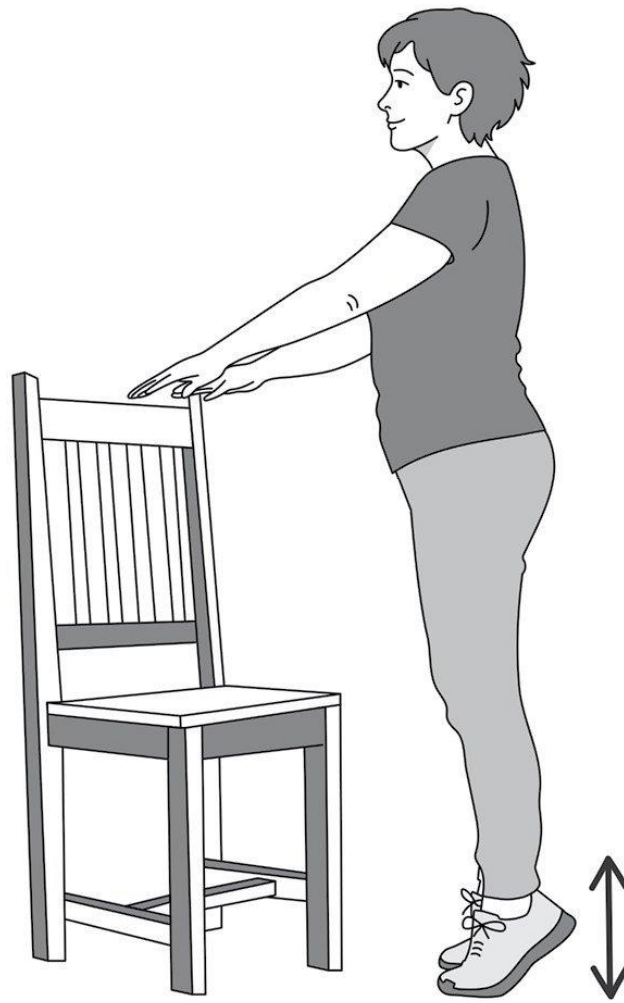


Figure 7.3. Standing heel raises create beneficial impact for the lower body.

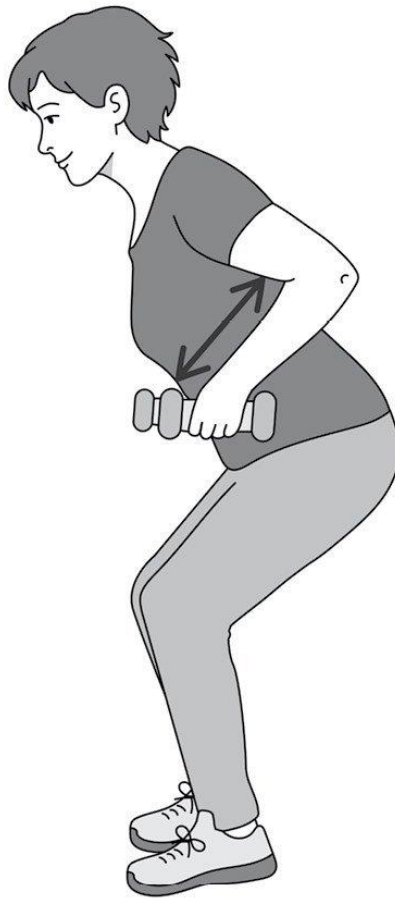


Figure 7.4. Dumbbell rows strengthen the back and improve posture.



Figure 7.5. Standing hip abduction targets the hip and improves balance.



Figure 11.1. Safe lifting requires keeping the spine straight while hinging at the hips.

Your First 30 Days

A printable action plan from *Osteoporosis: The Book I Wish I'd Had When I Was Diagnosed*. Chapter 12, set out as a checklist you can work through at your own pace. Nothing here is medical advice, and your own clinician may want you to do things in a different order.

Before you start

Finding out you have osteoporosis or osteopenia can feel overwhelming.

You might be wondering where to start, what to do first, and how to prioritize the many recommendations you're receiving.

This action plan will guide you through your first month, giving you concrete steps to take while you're still processing the diagnosis.

Important note: This timeline assumes you were diagnosed through screening (a DEXA scan without recent fractures).

If you're reading this because you recently fractured a bone, your first priority is healing.

Skip ahead to the "If You're Recovering from a Fracture" section for modified guidance that prioritizes recovery while still taking important first steps for your bone health.

Week 1: Information Gathering and Assessment

Days 1–3: Understand Your Diagnosis

- ☐ Request a copy of your complete DEXA scan results, including T-scores for all tested sites (spine, hip, forearm).
- ☐ Ask for your Z-score as well. If available, request your FRAX or FRAXplus score.
- ☐ Ask about TBS if it wasn't included.
- ☐ Write down any questions that come up as you review your results.

Days 4–7: Schedule Essential Appointments

- ☐ Make an appointment with your primary care doctor or endocrinologist to discuss treatment options (aim to do this within two to four weeks).
- ☐ Gather previous results if you've had relevant tests done (vitamin D level, calcium level, parathyroid hormone, thyroid function, complete blood count, kidney and liver function, bone turnover markers).
- ☐ If you haven't seen a dentist in the past year, schedule a dental checkup (important before starting certain bone medications).

Throughout Week 1: Start a Bone Health Journal

- ☐ Record your current height (measure against a wall).
- ☐ List all medications and supplements you're currently taking.
- ☐ Note any symptoms you've been experiencing (back pain, height loss, etc.).
- ☐ Document your family history of osteoporosis or fractures.
- ☐ Track questions to ask your doctor.

Week 2: Begin Safe Lifestyle Changes

Days 8–10: Assess Your Environment

- ☐ Walk through your home looking for fall hazards (refer to Chapter 11).
- ☐ Make a list of safety improvements needed.
- ☐ Start with quick fixes: remove throw rugs, improve lighting, clear clutter from walkways.
- ☐ Check that you have sturdy handrails where you need them.

Days 11–14: Start Gentle Movement

- ☐ If you aren't doing so already, begin walking daily, even if just for 10–15 minutes.
- ☐ Practice standing on one foot while holding on to a counter (work up to 30 seconds per side).
- ☐ Start noticing your posture throughout the day; aim to keep your tummy in and shoulders back.
- ☐ Avoid forward bending, twisting, or high-impact activities until you discuss exercise with your healthcare provider.

Throughout Week 2: Dietary Assessment

- ☐ Track what you eat for three to four days.
- ☐ Calculate your approximate daily calcium intake (see Chapter 8).
- ☐ Note how much caffeine, alcohol, and soda you typically consume.
- ☐ Identify one dietary change you can make this week (example: swap afternoon soda with water, or swap potato chips with almonds).

Week 3: Treatment Planning and Implementation

Days 15–17: Medical Appointments and Decisions

- ☐ Attend your doctor's appointment with your list of questions.
- ☐ First, discuss whether you need screening for other medical conditions that could be causing bone loss.
- ☐ Review your health history and medication list for possible “hidden thieves” (see Chapter 6).
- ☐ Expect, at minimum, some basic blood tests, such as a metabolic panel and a Vitamin D level to be ordered.
- ☐ Your healthcare provider should also consider tests for: thyroid or parathyroid disease, inflammatory disease, Celiac disease, diabetes, Cushing's disease, and multiple myeloma.
- ☐ After testing is done and any abnormal results addressed, ask about osteoporosis treatment: Is medication recommended for your situation?
- ☐ Could you try diet and exercise first, and if so, when will a follow-up DEXA be done?
- ☐ Discuss when to start, what side effects to watch for, how long you'll need to take the medication, and the monitoring schedule.

Days 18–21: Supplement Strategy

- ☐ Based on your bloodwork results, start appropriate supplements: calcium citrate (if dietary intake is insufficient), vitamin D (dosage based on your blood level), vitamin K2-MK7 (if recommended by your doctor), and magnesium (if not getting enough from diet).
- ☐ Consider collagen peptides (discuss with your doctor).
- ☐ Set up a system to remember daily supplements (phone alarm, pill organizer).
- ☐ Take calcium in divided doses (no more than 500 mg at once).
- ☐ Take calcium and vitamin D with food for better absorption.

Throughout Week 3: Medication Education

- ☐ If you're prescribed medication, read all patient information carefully and consult with your pharmacist.
- ☐ Note the best time to take it. Understand what to do if you miss a dose.
- ☐ Know what side effects warrant calling your doctor.
- ☐ Set up reminders to take medication correctly.

Week 4: Building Your Support System and Long-Term Plan

Days 22–24: Assemble Your Healthcare Team

- ☐ Identify who will manage your osteoporosis care (primary care doctor, endocrinologist, or another specialist).
- ☐ Ask for referrals if needed: physical therapist with osteoporosis expertise, registered dietitian, endocrinologist or rheumatologist with osteoporosis expertise.
- ☐ Schedule your next follow-up appointment (typically in three to six months).

Days 25–28: Exercise Program Development

- ☐ If possible, schedule an appointment with a physical therapist who specializes in osteoporosis.
- ☐ Discuss which exercises are safe and beneficial for your specific situation.
- ☐ Get guidance on proper form and progression.
- ☐ Create a realistic weekly exercise schedule (refer to Chapter 7).
- ☐ Start with two to three days of strength training and daily walking per week.

Days 29–30: Create Your Ongoing Management Plan

- ☐ Review what you've accomplished in the past month, and recalibrate your efforts as needed.
- ☐ Set three-month goals: exercise consistency target, dietary improvements, weight management goals (if applicable), fall-prevention measures completed.
- ☐ Mark your calendar for the following: next DEXA scan (typically in one to two years, but your doctor will advise), next medical appointment, regular medication refills, bloodwork follow-ups.

End of Month: Celebrate Your Progress

Take a moment to acknowledge that you've taken control of your bone health.

You've moved from shock and uncertainty to active management.

This is a significant accomplishment.

If You're Recovering from a Fracture

Your first 30 days look different, because healing takes priority. Here's your modified plan.

Week 1: Focus on Healing and Pain Management

- ☐ Follow your doctor's instructions for activity restrictions carefully.
- ☐ Take prescribed pain medication as directed; contact a pain management doctor if your pain is severe.
- ☐ Get adequate rest. Make sure you're eating enough protein and calories for healing.
- ☐ Start calcium and vitamin D supplementation (if not already taking).
- ☐ Ask about testing for other reasons for weakening of the bones.
- ☐ If you're younger (under 65) or have any history of cancer, it's important to ask your doctor whether more serious conditions (like cancer or multiple myeloma) could be causing your fractures.
- ☐ What to skip: Don't start an exercise program yet.
- ☐ Don't attempt balance exercises or strength training.
- ☐ Avoid assessing your home for fall hazards on your own (you may not be steady enough).

Week 2: Gentle Resumption of Activity

- ☐ Begin whatever gentle movement your doctor has cleared (this may be limited initially; you may be told to get moving right away or to rest as much as possible).
- ☐ Work with physical therapy if prescribed.
- ☐ Continue focusing on nutrition and healing.
- ☐ Start a bone health journal when you feel up to it.
- ☐ Don't rush recovery. Avoid any exercises that cause pain.

Week 3: Planning for Long-Term Management

- ☐ Schedule or attend appointments to discuss osteoporosis treatment.
- ☐ Begin to address one dietary change if you're feeling better.
- ☐ Review medications with your doctor (some may have contributed to your fracture risk).
- ☐ Ask about the timeline for more active bone-strengthening exercises.
- ☐ Don't be surprised if you start feeling down or discouraged; find someone to talk to about your feelings.

Week 4: Building Your Foundation

- ☐ If you're healing well, begin to implement home safety changes (with help).
- ☐ Start planning your exercise program for when you're fully healed.
- ☐ Focus on consistency with supplements and any prescribed medications.
- ☐ Set realistic expectations for your recovery timeline.

Important Reminders for Fracture Recovery

Healing varies widely by individual and fracture type.

Single vertebral fractures may heal in 6–12 weeks, but multiple vertebral fractures can take much longer (sometimes a year or more).

Hip fractures typically require three to six months to heal.

Ask your doctor about scheduling follow-up imaging to monitor healing progress.

Pain should gradually improve week by week, though the timeline varies, and you can expect pain to rebound a little bit when you start new activities.

Your activity level will increase slowly under medical supervision.

Some of the recommendations in this book (especially Chapter 7 on exercise) should be postponed until you're healed.

Be patient with yourself; recovering from a fracture while processing an osteoporosis diagnosis is genuinely difficult.

Beyond the First 30 Days

By the end of your first month, you've established the foundation.

Now, the focus shifts to maintaining consistency with medications and supplements, gradually building your exercise program, making ongoing dietary improvements, attending follow-up appointments, monitoring your progress, and staying educated about updates in bone health.

Remember, managing osteoporosis is a marathon, not a sprint.

The actions you take in these first 30 days set the stage for long-term bone health, but give yourself grace as you adapt to this new reality.

Some days will feel overwhelming. Other days, you'll feel empowered and in control. Both responses are normal.

You're not alone on this journey, and taking action, even small steps, makes a real difference in your bone health outcomes.

Keep moving forward, one day at a time.

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All 18 tables

The narration summarized these out loud. Here they are in full, with every number as it appears in the book.

Chapter 6: The Hidden Thieves

Medications and conditions that can quietly cost you bone.

Here's a guide to medications that may affect bone health. This list isn't exhaustive, and the presence of a medication here does not mean you should stop taking it. What's important is being aware of the potential bone effects, so you and your healthcare team can use appropriate monitoring (DEXA scans) and protective strategies (exercise and diet).

Medication Category	Examples	What to Know
Glucocorticoids/ corticosteroids	Prednisone (Deltasone), methylprednisolone (Medrol), dexamethasone (Decadron)	Even short courses can affect bones; long-term use causes dramatic bone loss
Aromatase inhibitors	Anastrozole (Arimidex), letrozole (Femara), exemestane (Aromasin)	Used for breast cancer; eliminate estrogen's bone protection
Androgen deprivation therapy	Leuprolide (Lupron), goserelin (Zoladex), degarelix (Firmagon)	Used for prostate cancer; eliminate testosterone's bone protection
Hormonal contraceptives (injectable)	Medroxyprogesterone (Depo-Provera)	FDA black box warning for bone loss; greatest risk with long-term use
SSRIs	Fluoxetine, sertraline, paroxetine, citalopram, escitalopram	Risk may increase with duration, especially after 5+ years
SNRIs	Venlafaxine, duloxetine	Similar possible risks to SSRIs
Anti-seizure medications	Phenytoin (Dilantin), carbamazepine (Tegretol), phenobarbital (Luminal), valproic acid (Depakote)	Enzyme-inducing drugs particularly affect bone metabolism
Proton pump inhibitors	Omeprazole, esomeprazole, lansoprazole, pantoprazole	Risk increases with long-term use (years, not months)
Excess thyroid hormone	Levothyroxine (Synthroid, Levoxyl) over-treatment	Too much accelerates bone turnover
Thiazolidinediones	Pioglitazone (Actos), rosiglitazone (for diabetes)	May increase fracture risk, particularly in women
Loop diuretics	Furosemide (Lasix)	May affect calcium excretion
Anticoagulants	Heparin (with long-term use)	Long-term use may affect bone
Methotrexate	High-dose or long-term use	May affect bone with extended use
Calcineurin inhibitors	Cyclosporine, tacrolimus (transplant medications)	Used after organ transplants; multiple mechanisms affect bone
Some antipsychotics	Various	May work through effects on a hormone called prolactin

Chapter 8: Nutrition and Bone Health

What to eat, in the amounts that actually matter.

In the United States, results are typically reported in ng/mL (nanograms per milliliter). Here's what those numbers mean:

Vitamin D Level	Status	Guidelines and Clinical Notes
Below 12 ng/mL (30 nmol/L)	Severe deficiency	Supplementation shows the clearest bone health benefits in this range.
12-20 ng/mL (30-50 nmol/L)	Deficient	Supplementation is strongly recommended to reach safe levels.
20-30 ng/mL (50-75 nmol/L)	Insufficient	May benefit from supplementation, especially for protecting bone health.
30-40 ng/mL (75-100 nmol/L)	Adequate	Generally considered adequate for maintaining bone health.
40-50 ng/mL (100-125 nmol/L)	Ideal	Considered the optimal target range for bone strength.
Above 50 ng/mL (125 nmol/L)	Potentially excessive	Provides no additional bone benefit and carries possible risks.

Dairy products are among the richest and most easily absorbed sources of calcium, and many are also fortified with vitamin D.

Food and Serving Size	Key Nutrients	Additional Notes
Cow's milk (1 cup)	Calcium: 300 mg; Protein: 8 g	Often fortified with vitamin D (check labels); budget-friendly option
Greek yogurt (1 cup)	Calcium: 250–300 mg; Protein: 15–20 g	Often contains vitamin D; provides probiotics for gut health; choose plain varieties to avoid added sugars
Cheese, hard varieties (e.g., cheddar, Parmesan) (1 oz)	Calcium: 200–250 mg; Protein: 7 g	Also provides vitamin K2; a little goes a long way due to concentrated nutrients
Cottage cheese (1 cup)	Calcium: 125–150 mg; Protein: 25 g	Lower in calcium than other dairy, but an excellent protein source; budget-friendly
Fortified plant milk (soy, almond, oat)(1 cup)	Calcium: 300–450 mg (when fortified) Vitamin D: often 100–140 IU	Soy milk provides 7–8 g protein, almond/oat milk typically 1–2 g; fortification varies: read labels and shake well before pouring

Fish, other animal proteins, and plant-based proteins all contribute to bone health through protein and various minerals. Aim for 0.8 grams per kg of your body weight (0.36 grams per pound of body weight) daily. This will end up being between 45 and 75 grams for most people.

Food and Serving Size	Key Nutrients	Additional Notes
Salmon, wild-caught (3 oz cooked)	Calcium: 180 mg (with bones in canned) Vitamin D: 570 IU; Protein: 17 g	One of the best natural sources of vitamin D; also provides omega-3s
Rainbow trout, farmed (3 oz)	Vitamin D: 600 IU; Protein: 20 g	One of the highest vitamin D contents among fish; mild flavor
Mackerel, canned (3 oz)	Vitamin D: 450–500 IU; Protein: 20 g	Often overlooked, but comparable to salmon for vitamin D; rich flavor
Sardines, canned with bones (3.75 oz can)	Calcium: 350–380 mg; Vitamin D: 150–200 IU; Protein: 23 g	Budget-friendly; the bones are soft and edible, and they provide most of the calcium
Tuna, canned (3 oz)	Vitamin D: 150 IU; Protein: 20 g	Budget-friendly; lower in calcium than sardines but still a good vitamin D source

Protein Sources: Other Animal Proteins

Food and Serving Size	Key Nutrients	Additional Notes
Chicken or turkey (3 oz cooked)	Protein: 25–27 g	Provides zinc and B vitamins that support bone health; versatile and budget-friendly
Eggs (1 large)	Vitamin D: 40 IU (in yolk) Protein: 6 g	Also provides vitamin K2; budget-friendly; the vitamin D is in the yolk, not the white
Lean beef (3 oz cooked)	Protein: 25 g	Provides zinc and vitamin K2; choose grass-fed when possible for higher omega-3 content
Pork tenderloin or loin (3 oz cooked)	Protein: 22–24 g	Provides thiamin (vitamin B1) and zinc; choose lean cuts to minimize saturated fat

Protein Sources: Plant-Based

Food and Serving Size	Key Nutrients	Additional Notes
Tofu, calcium-set (½ cup)	Calcium: 250–750 mg; Protein: 10 g	Only calcium-set tofu provides significant calcium; check ingredients for “calcium sulfate” or “calcium chloride”
Tempeh (½ cup)	Calcium: 90 mg; Protein: 15 g	Provides probiotics and is less processed than many meat alternatives
Edamame (1 cup)	Calcium: 100 mg; Protein: 17 g	Also provides magnesium and other minerals
Lentils and beans (1 cup cooked)	Calcium: 40–80 mg; Protein: 15–18 g	Also provide magnesium, zinc, and other trace minerals; budget-friendly; legumes provide multiple bone-supporting nutrients

Dark leafy greens can be excellent calcium sources, but absorption varies significantly depending on their oxalate content. (Oxalate is a natural compound found in some vegetables.)

Food and Serving Size	Key Nutrients	Additional Notes
Collard greens, cooked (1 cup)	Calcium: 270 mg; Vitamin K: 770 mcg	Excellent absorption (about 50% vs. 30% from dairy); also provides magnesium; budget-friendly
Bok choy, cooked (1 cup)	Calcium: 160 mg; Vitamin K: 58 mcg	Good calcium absorption; low-calorie and versatile
Kale, cooked (1 cup)	Calcium: 180 mg; Vitamin K: 1,060 mcg	Good calcium absorption; also provides magnesium; available fresh or frozen
Mustard greens, cooked (1 cup)	Calcium: 165 mg; Vitamin K: 830 mcg	Peppery flavor; nutritional powerhouse that also provides vitamin A and C
Turnip greens, cooked (1 cup)	Calcium: 200 mg; Vitamin K: 530 mcg	Often overlooked, but an excellent choice
Okra, cooked (1 cup)	Calcium: 120 mg; Magnesium: 90 mg	Also provides vitamin K and folate; excellent absorption
Broccoli, cooked (1 cup)	Calcium: 60 mg; Vitamin K: 220 mcg	Good calcium absorption; also provides vitamin C and other antioxidants

Vegetables and Leafy Greens: Lower Absorption (High Oxalate)

Food and Serving Size	Key Nutrients	Additional Notes
Spinach (1 cup cooked)	Calcium: 245 mg (only ~5% absorbed)	High oxalate content limits calcium absorption; still nutritious for other reasons (iron, vitamins), but not a reliable calcium source; same issue with Swiss chard and beet greens

Almonds, almond butter, chia seeds, sesame seeds/tahini, and Brazil nuts all provide calcium along with other bone-supporting nutrients.

Food and Serving Size	Key Nutrients	Additional Notes
Almonds (1 oz, about 23 nuts)	Calcium: 75 mg; Protein: 6 g; Magnesium: 75 mg	Vitamin E and healthy fats; portable and shelf-stable
Almond butter (2 tablespoons)	Calcium: 85 mg; Protein: 7 g; Magnesium: 90 mg	Now commonly available and a great substitute for peanut butter
Pumpkin seeds (pepitas) (1 oz)	Magnesium: 150 mg; Zinc: 2 mg; Protein: 8 g	One of the best plant sources of magnesium; also provides iron and healthy fats
Walnuts (1 oz)	Omega-3s (ALA): 2.5 g; Magnesium: 45 mg	Excellent for reducing inflammation, which supports bone health
Chia seeds (2 tablespoons)	Calcium: 180 mg; Protein: 4 g; Omega-3 fatty acids (ALA): 5 g	Can be added to smoothies, yogurt, or oatmeal
Flaxseeds, ground (2 tbsp)	Omega-3s (ALA): 3 g; Magnesium: 55 mg	Must be ground for body to absorb nutrients; easy to add to oatmeal or yogurt
Sesame seeds/tahini (2 tbsp)	Calcium: 130 mg (whole) or 65 mg (tahini)	Also provide magnesium and zinc; unhulled sesame seeds have much more calcium than hulled
Brazil nuts (1 oz, about 6 nuts)	Selenium: ~544 mcg (989% of the daily recommended value), one of the richest food sources of selenium on Earth Magnesium: 105 mg	Also provide calcium and protein; limit to 1–2 nuts daily due to very high selenium content

Fortified Foods and Whole Grains

Food and Serving Size	Key Nutrients	Additional Notes
Fortified breakfast cereals	250–1,000 mg calcium per serving (varies)	Often fortified with vitamin D; check labels, as fortification varies dramatically by brand
Fortified orange juice (1 cup)	Calcium: 300–350 mg; Vitamin D: often 100 IU	Vitamin C enhances calcium absorption; shake well before pouring, as calcium settles at the bottom
Oatmeal, instant fortified (1 pkt)	Calcium: 100–150 mg	Also provides magnesium; choose plain varieties and add your own fruit to control sugar
Amaranth, cooked (1 cup)	Calcium: 116 mg; Magnesium: 160 mg	An ancient grain (technically a seed) with unusually high calcium; nutty flavor
Teff, cooked (1 cup)	Calcium: 123 mg; Magnesium: 126 mg	The grain used in Ethiopian injera bread; excellent calcium source for a grain
Whole grain bread, fortified (1 slice)	Calcium: 50–150 mg	Fortification varies, so check labels; also provides other nutrients and fiber

Other Bone-Supporting Foods

Food and Serving Size	Key Nutrients	Additional Notes
Prunes/dried plums (5–6 prunes)	Vitamin K, boron, and trace minerals	Research shows prunes may help prevent bone loss; they also support digestive health
Blackstrap molasses (1 tbsp)	Calcium: 200 mg; Magnesium: 40 mg; Iron: 3.5 mg	A traditional bone-builder; significantly higher nutrients than regular molasses; strong, bitter-sweet flavor
Kiwi (1 medium)	Vitamin K: 30 mcg; Vitamin C: 64 mg	Vitamin C is essential for collagen formation in bone; very high vitamin C content per calorie
Figs, dried (¼ cup)	Calcium: 60 mg	Also provide magnesium and potassium; natural sweetness for snacks or cooking
Mushrooms (UV exposed, 3 oz)	Vitamin D: up to 400 IU	Must be UV exposed; most commercial mushrooms have minimal vitamin D unless specifically treated, so check the packaging
Natto (fermented soybeans) (3 oz)	Vitamin K2: 850 mcg; Calcium: 100 mg; Protein: 15 g	Acquired taste, but one of the richest food sources of vitamin K2

While focusing on adding bone-healthy foods is most important, it's also worth being aware of foods and beverages that can undermine your bone health efforts.

Food/Beverage to Limit	Impact on Bone Health	Guidelines and Better Alternatives
Excessive caffeine	>3–4 cups daily can increase calcium excretion	If you drink coffee heavily, ensure adequate calcium intake and consider cutting back gradually
Cola and soft drinks (including diet)	Phosphoric acid can disrupt calcium–phosphorus balance	Better choices: water, sparkling water with lemon, or herbal tea
Ultra-processed foods	Phosphate additives in baked goods, processed meats, and frozen meals may negatively affect calcium balance	Check labels for “phosphate” or “phosphoric acid”; choose minimally processed foods
Excessive alcohol	>2–3 drinks daily interferes with bone formation	Moderate consumption (1 drink/day for women, 1–2 for men) appears safe
Excessive salt	High sodium increases calcium excretion	Mainly a concern with processed foods; limit ultra-processed items

Chapter 9: Medication Choices

The osteoporosis medications, side by side.

Understanding the side effects is reassuring for most people. Serious side effects are uncommon. The most common issues involve stomach upset with oral forms of the medication and flu-like symptoms after IV infusions. Two rare but serious concerns with long-term use are osteonecrosis (death of bone) inside the jaw, which is more common in people receiving high-dose IV bisphosphonates for cancer or those who undergo teeth extraction, and atypical femur fractures, which are unusual breaks in the thigh bone. Both are very rare in people taking standard doses for osteoporosis. Osteonecrosis of the jaw has been reported in 1 in 100,000 to 1 in 1,000 patients, with the higher rates seen in long-term users. This helps to explain why some doctors recommend a “drug holiday” after several years of treatment.

Medication and Usage	Mechanism and Side Effects	Important Notes
Alendronate (Fosamax, Fosavance, Vantavo) Pill: Daily or weekly Cost: Low	Action: Slows osteoclast activity; preserves existing bone. Side effects: Heartburn, stomach upset, esophageal irritation. Rare: jaw problems, unusual thigh fractures.	Generic available. Must take on empty stomach and stay upright for 30–60 minutes. Reduces both vertebral and hip fractures by 40–50%.
Risedronate (Actonel, Atelvia) Pill: Daily, weekly, or monthly Cost: Low	Action: Slows osteoclast activity. Side effects: Like alendronate, but may have slightly less gastrointestinal (GI) upset.	Generic available; flexible dosing. Similar efficacy and administration rules to alendronate.
Ibandronate (Bonviva, Bondronat, Bondenza) Pill: Monthly IV: Every 3 months Cost: Low to medium	Action: Slows osteoclast activity. Side effects: Pill: Similar GI effects to alendronate. IV: Flu-like symptoms for 1–3 days post-infusion.	Less frequent dosing. IV form avoids GI side effects. Proven benefit mainly for spine fractures.
Zoledronic acid (Reclast, Aclasta, Zometa) IV infusion: Once yearly Cost: Medium	Action: Slows osteoclast activity. Side effects: Flu-like symptoms (fever, aches) 1–3 days after first dose. Rare: kidney problems.	Convenient yearly dosing. No GI side effects. Requires adequate hydration and kidney monitoring. Very effective for all fracture types, with up to a 70% reduction in vertebral fractures.

The potential downsides of denosumab include the potential for low blood calcium levels (after all, bone breakdown is the body’s way of rapidly increasing blood calcium levels), especially in people with kidney problems, and the need for careful planning around any treatment discontinuation. Stopping denosumab suddenly can lead to rapid bone loss, so transitions to other medications require careful timing and monitoring.

Medication and Usage	Mechanism and Side Effects	Important Notes
Denosumab (Prolia, Xgeva) Injection: Every 6 months Cost: Medium to high	Action: Blocks signals that activate osteoclasts. Side effects: Low calcium, skin infections, back/muscle pain. Rare: jaw problems, unusual thigh fractures.	No stomach issues. Excellent bone density improvements. Cannot stop abruptly (rapid bone loss occurs). Must transition to another drug if stopping.

I’ll be taking teriparatide for another year or two from the time of this writing. Anabolic medications used to have a firm two-year lifetime use restriction based on early studies in rats, but this limitation was removed by the FDA in November 2020 after real-world evidence showed no increased cancer risk in humans. However, most

treatment courses still last 18–24 months, as this represents the time frame for optimal bone-building benefit. The important consideration is that anabolic medications work best when followed by antiresorptive therapy to maintain the bone gains achieved. Without this follow-up treatment, much of the benefit can be lost over time.

Medication and Usage	Mechanism and Side Effects	Important Notes
Teriparatide (Forteo, Forsteo) Daily self-injection Cost: Very high	Action: Stimulates osteoblasts; actively builds new bone. Side effects: Nausea, leg cramps, dizziness. Temporary high calcium levels.	Most potent bone-builder. Increases spine density significantly. Treatment typically 18–24 months. Must follow with antiresorptive to maintain gains. Available as a generic with a modest reduction in cost.
Abaloparatide (Tymlos, Eladynos) Daily self-injection Cost: Very high	Action: Stimulates osteoblasts; actively builds new bone. Side effects: Dizziness, nausea, headache, palpitations.	Like teriparatide. Treatment limited to 2 years. Must follow with antiresorptive to maintain gains.
Romosozumab (Evenity) Injection: Monthly for 12 months Cost: Very high	Action: Dual-action: increases bone formation and decreases resorption. Side effects: Joint pain, headache. Caution: May increase risk of cardiac events.	Rapid density gains. Limited to 12 months. Reserved for those with very high risk. Not for recent heart attack/stroke patients. Must follow with antiresorptive.

The downsides include hot flashes, which can be problematic for women still experiencing menopausal symptoms, and an increased risk of blood clots (similar to estrogen therapy). Leg cramps are another common complaint; some women find these bothersome enough that they discontinue treatment.

Medication and Usage	Mechanism and Side Effects	Important Notes
Raloxifene (Evista) Pill: Daily Cost: Low to medium	Action: Acts like estrogen in bones; blocks estrogen in breast tissues. Side effects: Hot flashes, leg cramps, increased blood clot risk.	Reduces vertebral fractures ~35% (no proven hip benefit). Bonus: Reduces breast cancer risk ~35%. Not for women with hot flashes.

If you’re considering HRT, the conversation with your doctor should include your individual risk factors for breast cancer, heart disease, and blood clots, as well as the severity of your menopausal symptoms and your fracture risk. The decision is highly personal and needs you to weigh multiple factors unique to your situation.

Medication and Usage	Mechanism and Side Effects	Important Notes
Hormone replacement therapy (HRT) Pill, patch, or cream: Daily Cost: Low	Action: Replaces estrogen to maintain bone balance. Side effects: Breast tenderness, bloating. Risks: Clots, stroke, heart disease.	Reduces all fractures by 20–40%. Only drug effective at all risk levels. Benefits stop when medication stops. Best for women with menopausal symptoms.

If you’re currently taking calcitonin for osteoporosis, it’s worth discussing with your doctor whether more effective alternatives might be appropriate for you.

Medication and Usage	Mechanism and Side Effects	Important Notes
Calcitonin Nasal spray or injection Cost: Low	Action: Temporarily inhibits osteoclasts. Side effects: Nasal irritation, nausea, flushing.	Rarely used. Modest spine benefit only. Small potential cancer risk with long-term use. Less effective than other options.

The rest of the figures

Bone anatomy, T-scores, how fractures happen, and the MRI of my own spine.

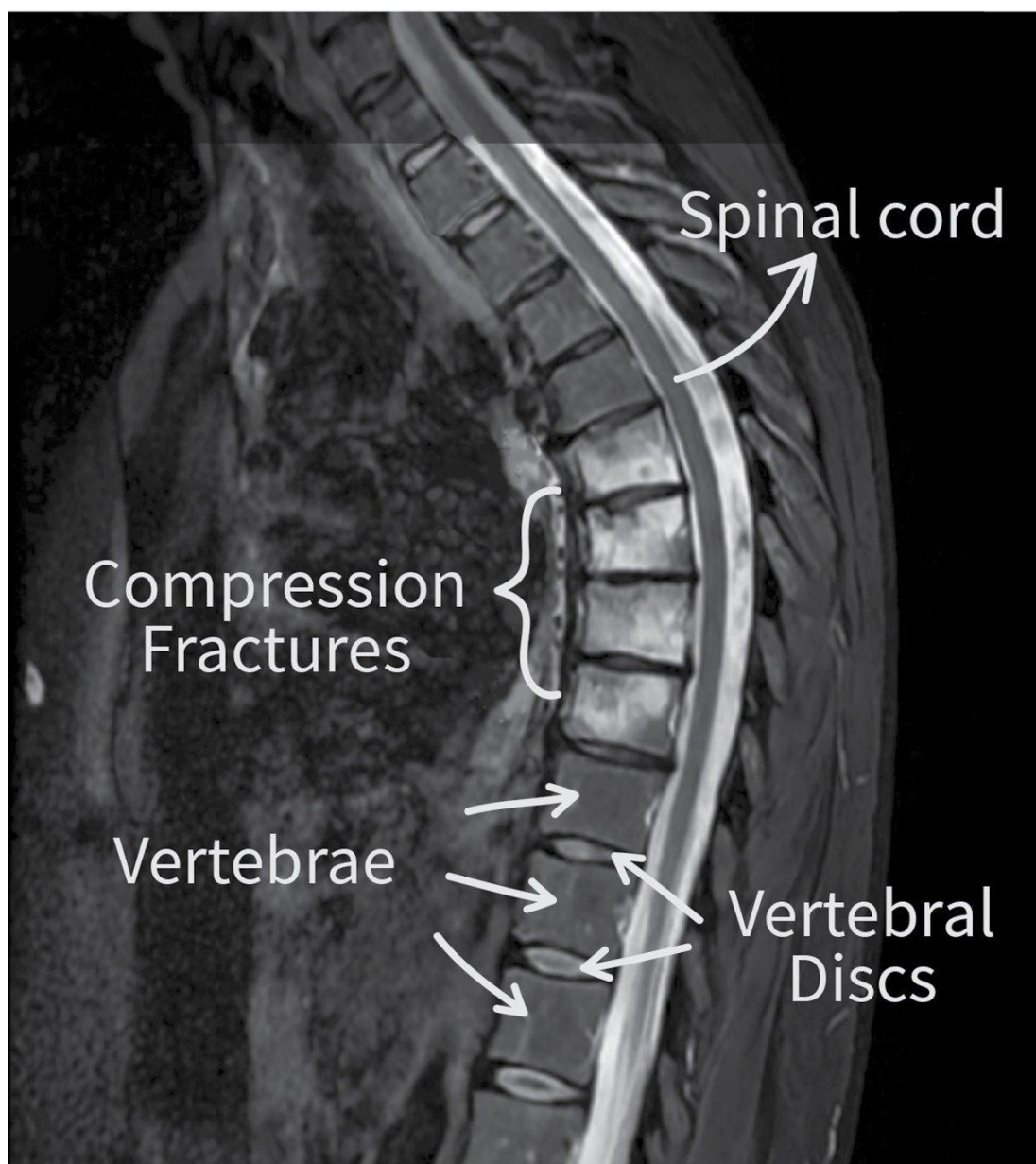


Figure 1.1. MRI of my thoracic spine (February 2025). The bracket highlights compression fractures at T6–T9 with bone marrow edema (the bright signal indicates acute injury). These fractures weren't visible on plain X-ray.

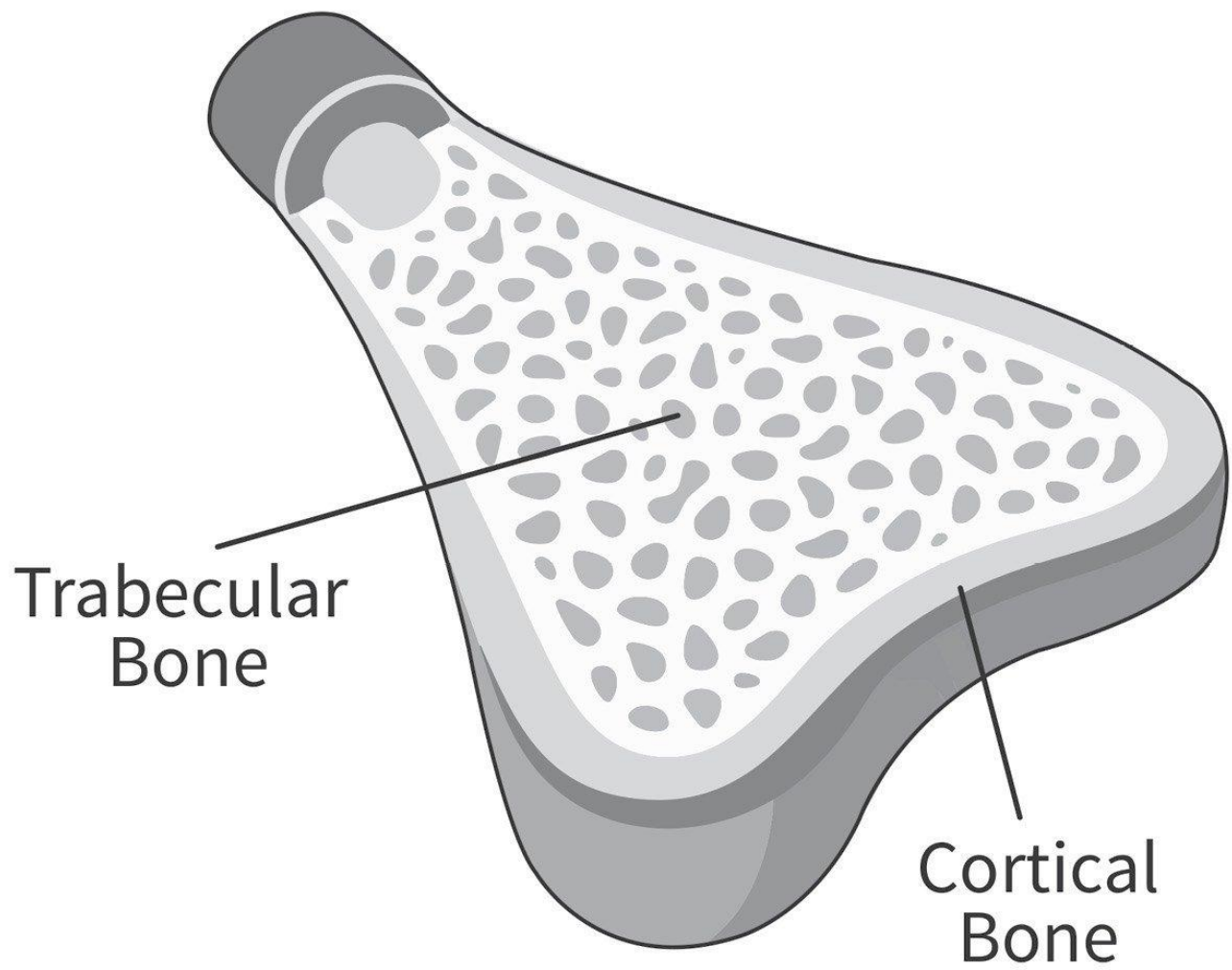


Figure 3.1. Cross-section of bone showing the dense outer cortical layer and the spongy inner trabecular (cancellous) bone.

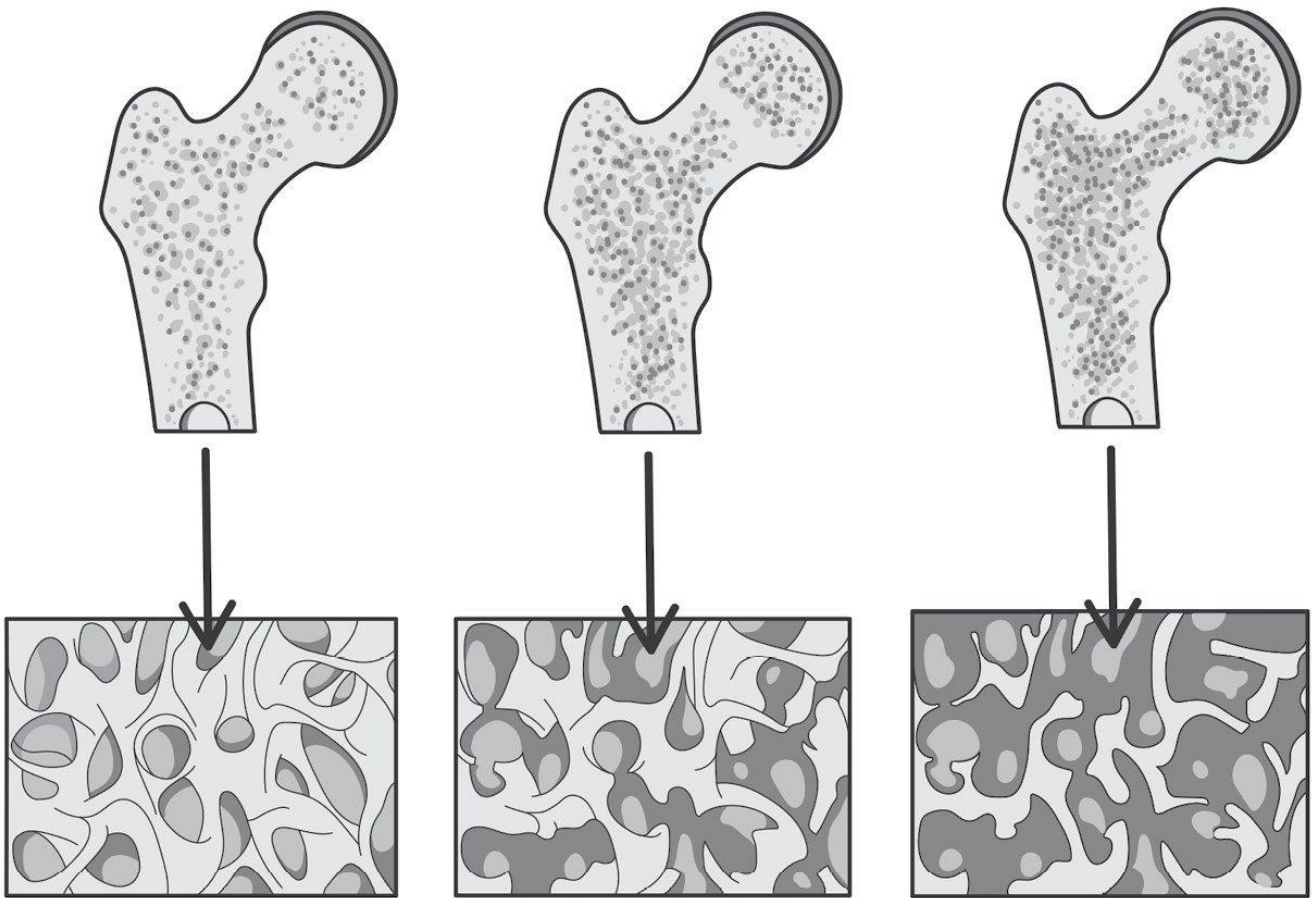


Figure 3.2. Comparison of the progression from normal bone density (left) to osteoporotic bone (right). Trabecular bone loses its dense honeycomb structure, leaving bones fragile and prone to fracture.

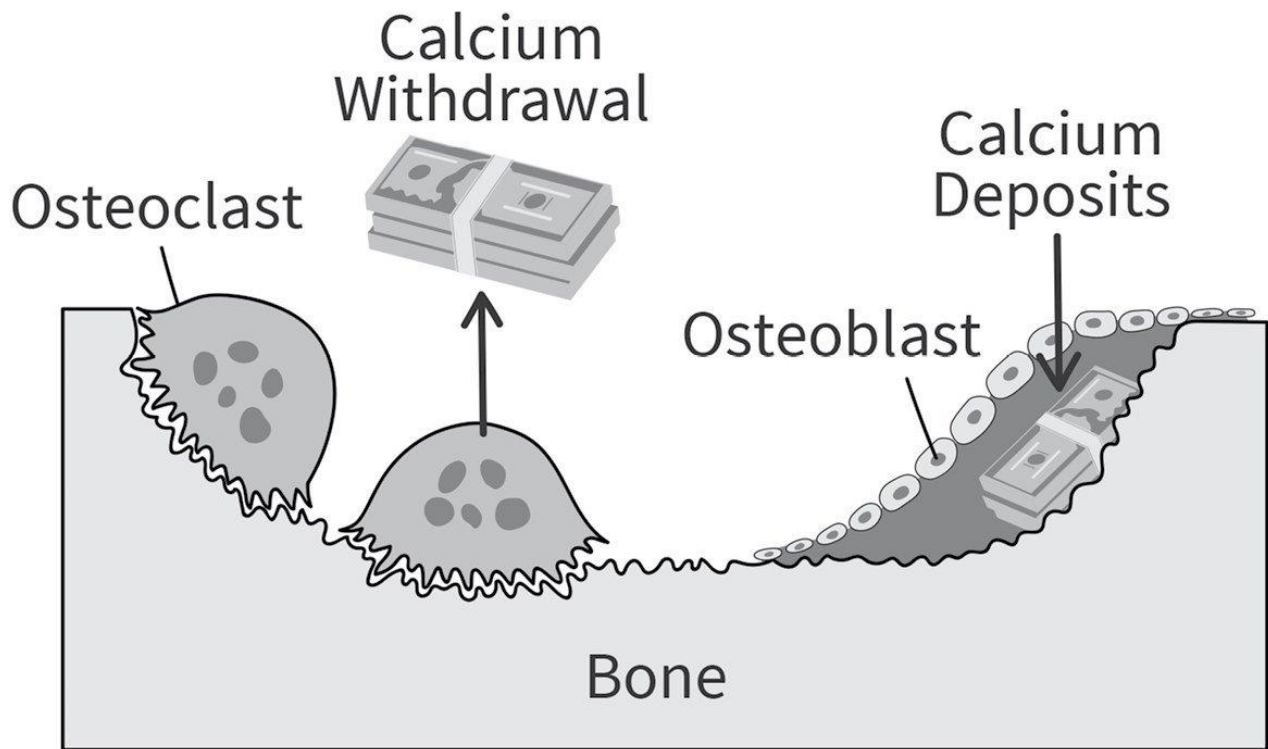


Figure 3.3. The “bone bank” metaphor. Osteoclasts make withdrawals by breaking down bone, while osteoblasts make deposits by building new bone matrix.

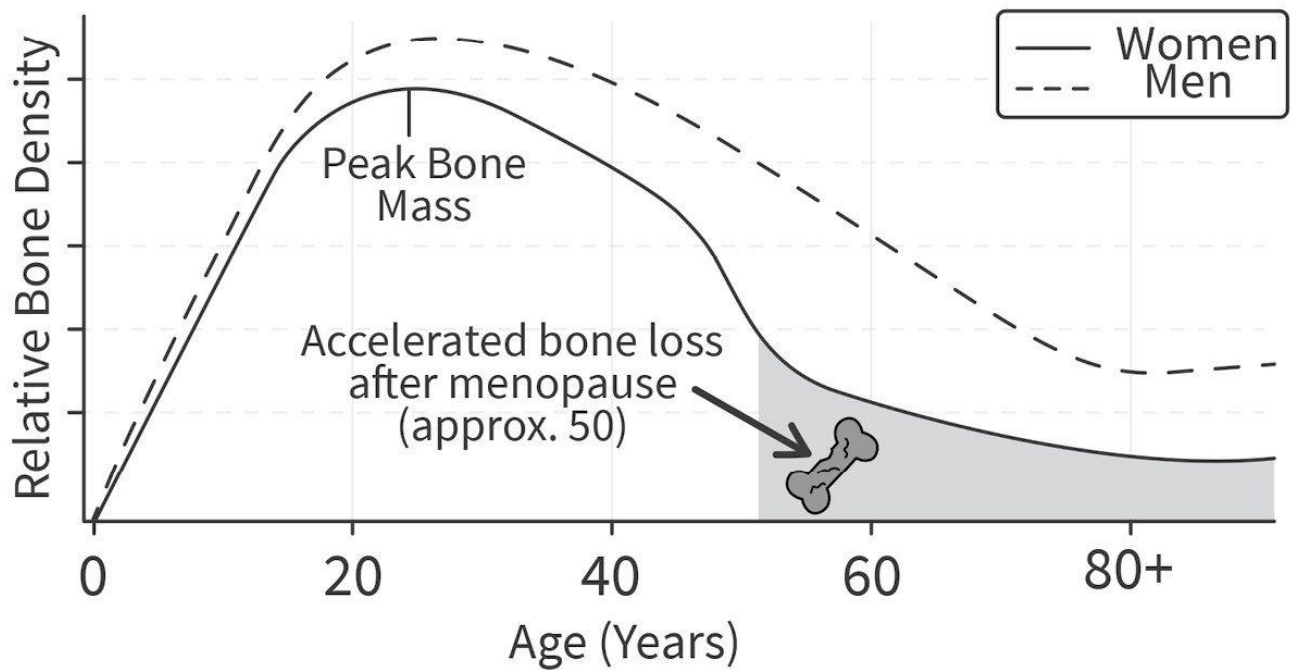
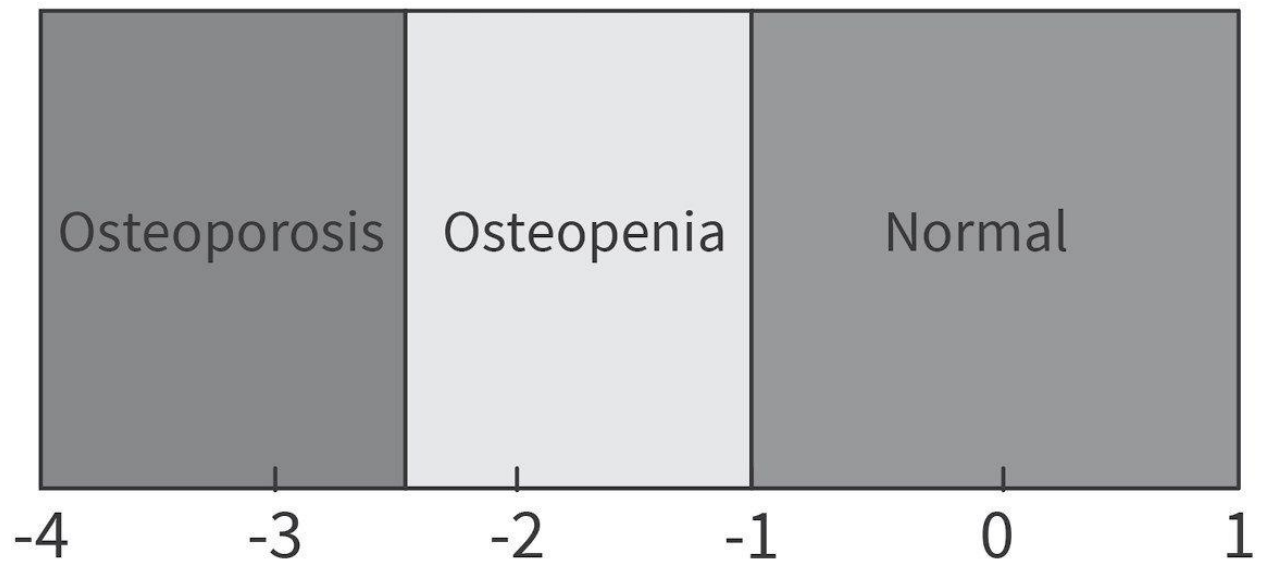


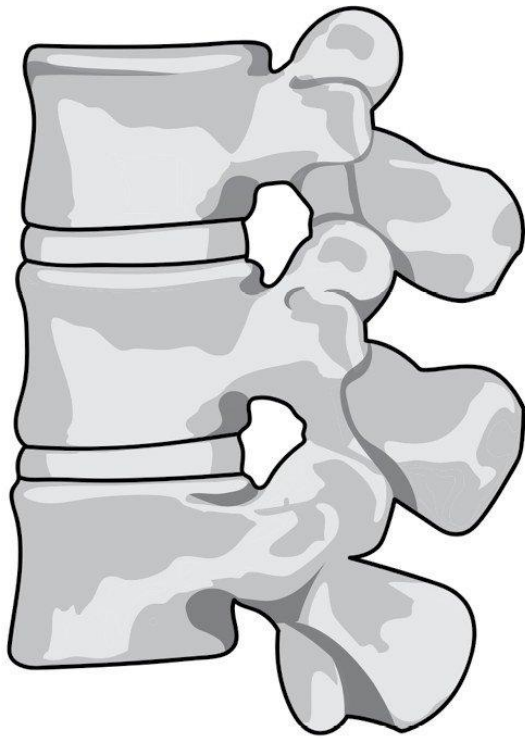
Figure 3.4. Bone density changes over the lifespan. Note the higher peak bone mass in men and the rapid decline in women following menopause.



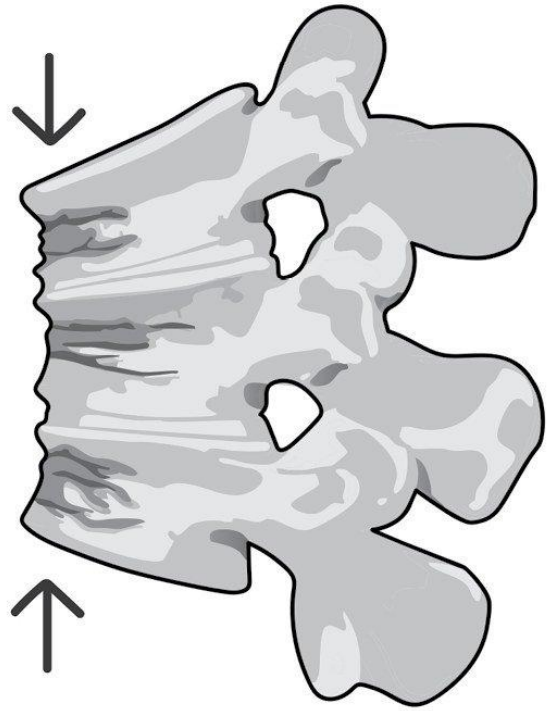
T-score

Note: Osteopenia plus a low-impact fracture is diagnosed as osteoporosis.

Figure 4.1. Understanding T-scores and their diagnostic categories.



Normal
Spine



Compression
Fractures

Figure 5.1. Vertebral compression fracture. The arrows show the compressive forces that cause a weakened vertebra to collapse into a wedge shape, which can lead to height loss and spinal curvature.

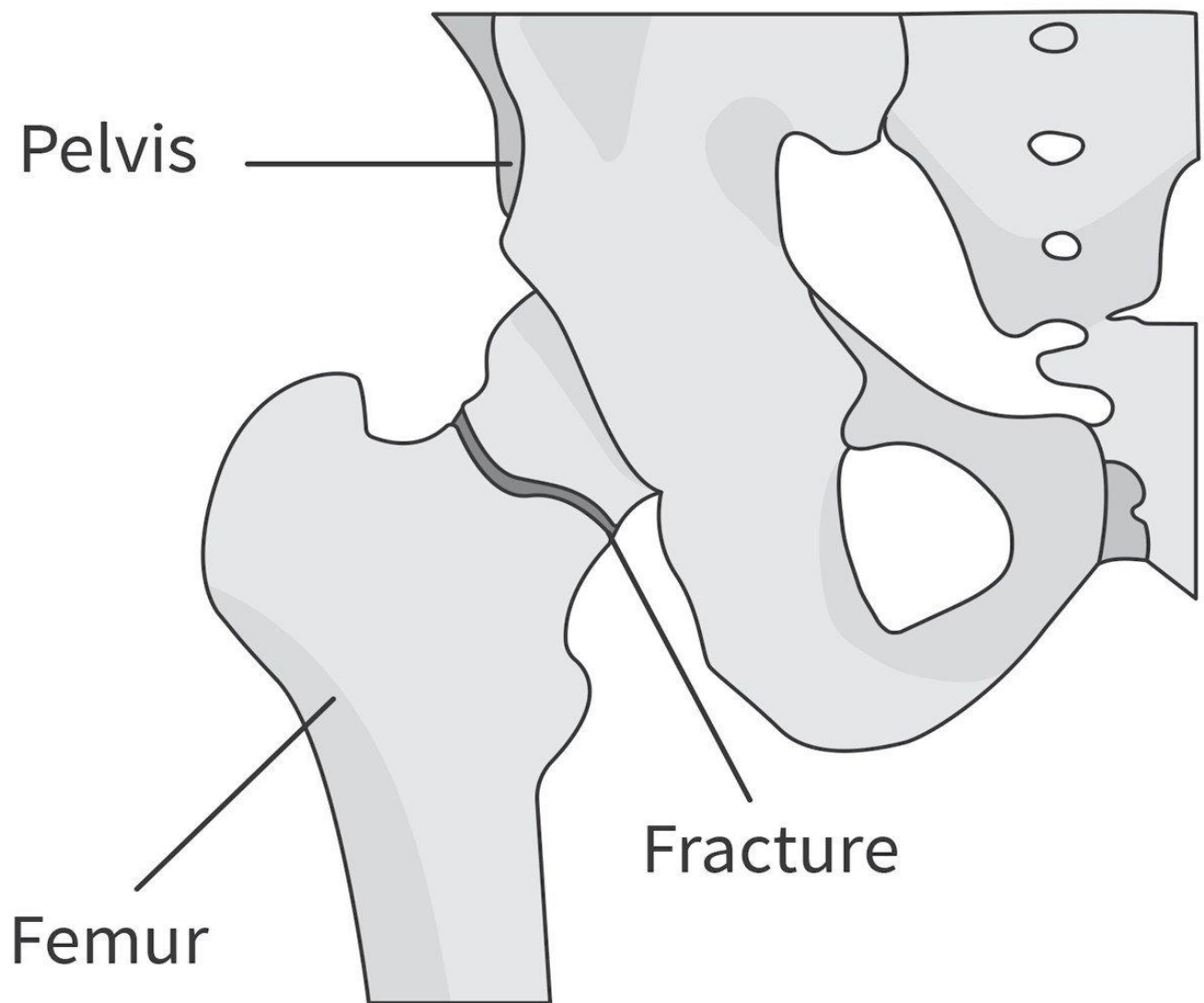


Figure 5.2. Hip fracture occurring at the femoral neck, the most common location for osteoporotic hip fractures.

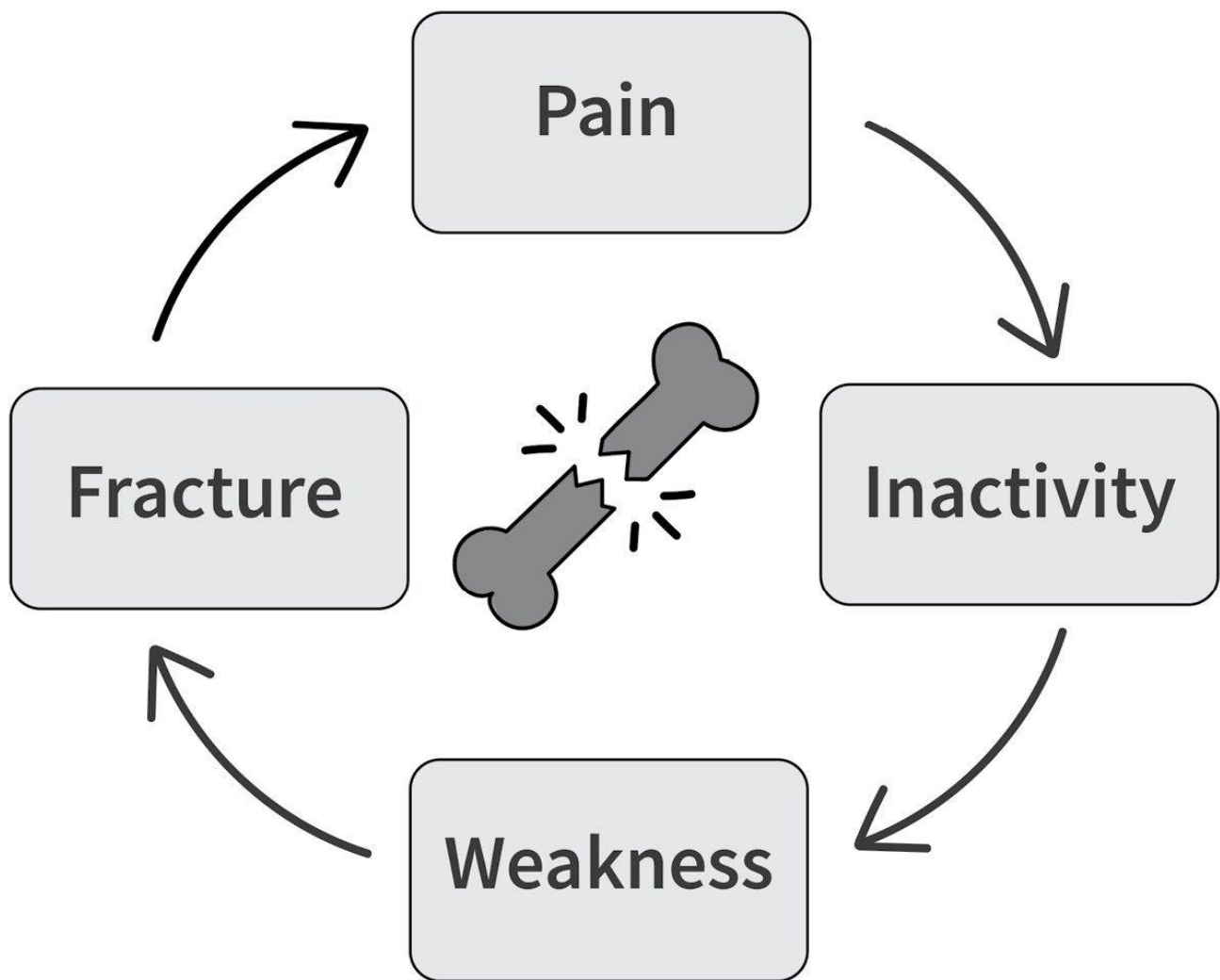


Figure 5.3. Fracture pain can lead to inactivity and weakness, leading to another fracture.

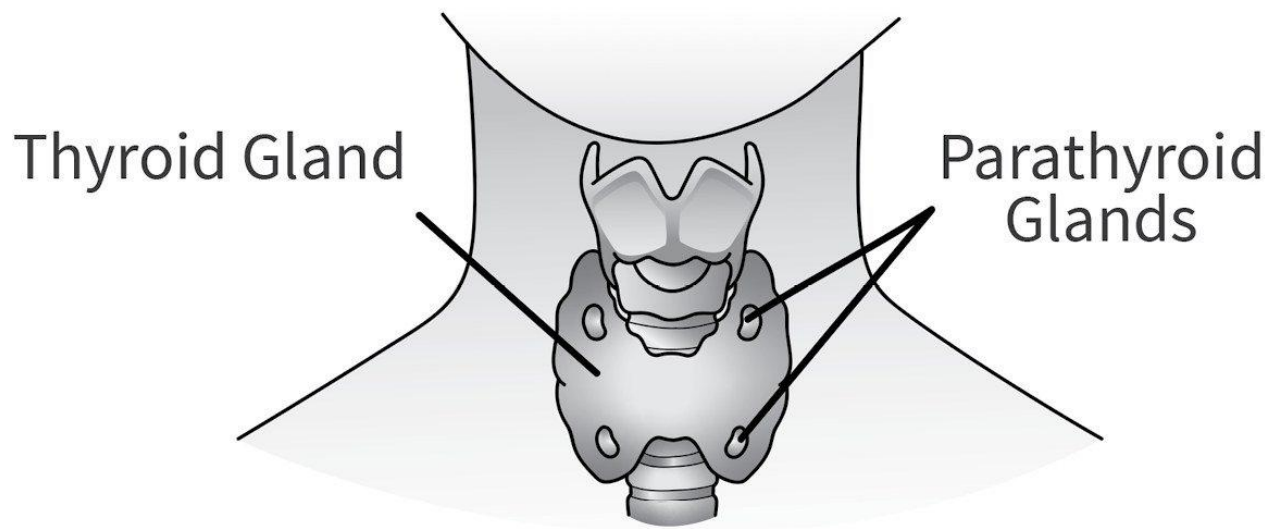


Figure 6.1. The location of the parathyroid glands.

Glossary, with pronunciations

Some of these words are hard to say out loud. The guides in parentheses show where the stress falls.

Abaloparatide (a-bal-oh-PAR-ah-tide)

An anabolic (bone-building) medication (brand name Tymlos) given as a daily injection.

Alendronate (ah-LEN-droe-nate)

A bisphosphonate medication (brand name Fosamax) taken orally to treat osteoporosis.

Bisphosphonates (bis-FOS-foe-nates)

A class of medications that slow bone loss by inhibiting osteoclast activity.

Calcitonin (kal-sih-TOE-nin)

A naturally occurring hormone that slows bone breakdown. Synthetic salmon calcitonin was once prescribed as a nasal spray for osteoporosis, but it has largely been replaced by more effective medications.

Cancellous bone (KAN-sell-us)

The spongy, honeycomb-like interior bone tissue; also called trabecular bone.

Denosumab (den-OH-sue-mab)

A biologic medication (brand name Prolia) given by injection every six months to slow bone loss.

Estrogen (ES-troh-jen)

A hormone that plays a key role in maintaining bone density. Estrogen therapy, sometimes combined with other agents, is used to prevent bone loss in postmenopausal women.

Hyperparathyroidism (high-per-para-THY-royd-ism)

A condition in which one or more parathyroid glands produce too much parathyroid hormone, leading to increased bone breakdown.

Ibandronate (eye-BAN-droe-nate)

A bisphosphonate medication (brand name Boniva) available in monthly oral or quarterly intravenous forms.

Kyphoplasty (KYE-foe-plas-tee)

A minimally invasive procedure to treat vertebral compression fractures by inflating a balloon in the fractured vertebra and filling it with bone cement.

Kyphosis (kye-FOE-sis)

An exaggerated forward curvature of the upper spine, sometimes called a “dowager’s hump.”

Osteoblasts (OS-tee-oh-blasts)

Bone-building cells that create new bone tissue.

Osteoclasts (OS-tee-oh-clasts)

Bone-resorbing cells that break down old or damaged bone tissue.

Osteopenia (os-tee-oh-PEE-nee-ah)

A condition of mildly reduced bone density, between normal bone density and osteoporosis.

Osteoporosis (os-tee-oh-pore-OH-sis)

A condition in which bones become weak and brittle due to loss of bone density and mass.

Physiatrist (fiz-EYE-ah-trist)

A physician specializing in physical medicine and rehabilitation, trained to diagnose and treat conditions affecting the muscles, bones, nerves, and brain without surgery.

Raloxifene (ral-OX-ih-feen)

A selective estrogen receptor modulator (SERM) used to prevent and treat osteoporosis in postmenopausal women.

Risedronate (rih-SEH-droe-nate)

A bisphosphonate medication (brand name Actonel) taken orally to treat osteoporosis.

Romosozumab (roe-moe-SOE-zoo-mab)

A dual-action medication (brand name Evenity) given as a monthly injection that both builds new bone and slows bone loss.

Teriparatide (ter-ih-PAR-ah-tide)

An anabolic (bone-building) medication (brand name Forteo) given as a daily injection.

Thoracic (thor-AS-ik)

Relating to the chest region of the spine (the 12 vertebrae between the neck and lower back).

Trabecular bone (trah-BEK-you-lar)

Seecancellous bone.

Vertebrae (VER-teh-bray)

The individual bones that make up the spinal column.

Vertebroplasty (ver-TEE-bro-plas-tee)

A minimally invasive procedure to treat vertebral compression fractures by injecting bone cement directly into the fractured vertebra to stabilize it. Unlike kyphoplasty, no balloon is used to restore height before the cement is injected.

Zoledronic acid (zoe-leh-DRON-ik)

A bisphosphonate medication (brand name Reclast) given as an annual intravenous infusion.

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Every endnote from the book, in order, with links where the source is online.

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